

CSC 580 - Artificial Intelligence – 2

Default Project -1 World Model

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References: Gemini, Github

Word Count – 3744

Introduction

This report involves the design and implementation of a Dreamer V1-style world model in the highway-v0 environment in the Gymnasium Highway-Env library, its training, and evaluation. The project, called Dream the Road -- Tiny Dreamer for Highway-env, is an example of how a reinforcement learning agent can be trained to navigate a multi-lane highway under continuous control, using an initial training of a latent dynamics model of the world using raw pixel observations, followed by policy training entirely in imagined latent-space trajectories.

Sample efficiency is one of the most critical issues of RL solved by model-based reinforcement learning (MBRL). Learning of competent policies in model-free approaches like PPO or SAC takes millions of interactions with the environment. World models on the contrary are trained to learn a low-dimensional predictive control of the environment dynamics, allowing the agent to dream thousands of simulated trajectories at no actual cost. Dreamer framework, proposed by Hafner et al. (2020), can be seen as an operationalization of this concept, which uses Recurrent State-Space Model (RSSM) and actor-critic architecture, which is trained using only imagined rollouts.

The highway-v0 setting offers a very attractive testbed to this method: the visual features of the surroundings (road, lane markings, vehicles) are relatively low-dimensional, but the dynamics are non-trivial (because of the stochastic dynamics of surrounding traffic governed by the Intelligent Driver Model (IDM)). Another complexity that exists over discrete-action environments is the continuous action space (steering and acceleration in $[-1, 1]$). Our implementation shows the entire Dreamer V1 pipeline, which includes data collection at pixels all the way to the policy deployed, at 0% collision and mean evaluation score of 63.09 on 10 test episodes.

Background

World Models in RL

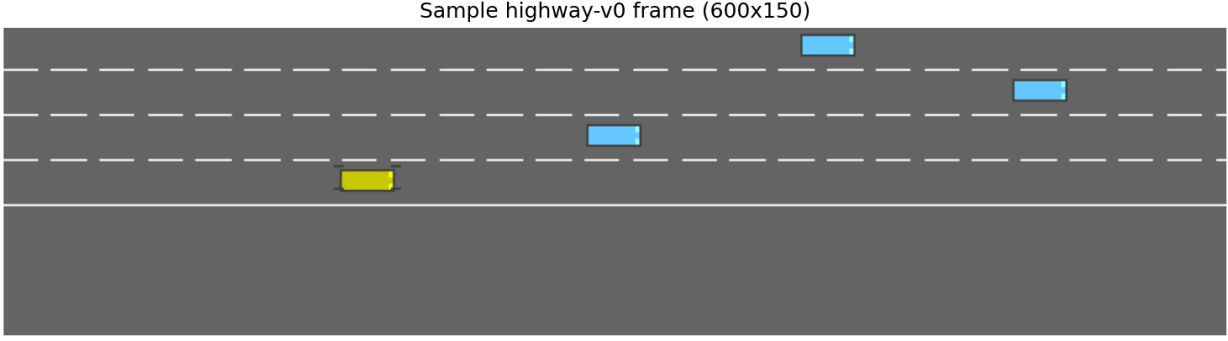
World models are learned with a forward dynamics function which forecasts future states and rewards provided current states and actions. Ha and Schmidhuber (2018) showed that agents were able to learn effective policies with training in a learned world model, and then apply the behaviors to the real world. They took a perceptual Variational Autoencoder (VAE) and a dynamics Mixture Density Network-RNN. This paradigm was built upon by Dreamer V1 (Hafner et al., 2020) which added the RSSM; a hybrid model composed of deterministic recurrent dynamics (GRU) and stochastic latent variables and by training the policy by backpropagating the value gradients through the learned dynamics instead of having to perform separate simulation rollouts.

The RSSM Architecture

Recurrent State-Space Model has two complementary representations of the state. The deterministic component h_t , which is updated by a GRU cell, represents the temporal structure and long-range dependencies in environment. Stochastic component z_t , which is a sample of a learned Gaussian distribution, represents the underlying uncertainty, multi-modality of future states. z_t is sampled during training out of the posterior distribution $q(z_t | h_t, x_t)$ which is allowed access to actual observation x_t . In imagination, z_t is sampled based on the previous $p(z_t | h_t)$ which has to predict the latent state without the knowledge of the surrounding environment. KL divergence between the posterior and prior is minimized in the course of training to make sure that the prior is a true predictor of the beliefs of the posterior.

Highway-Env Environment

The highway-v0 setup approximates a multi-lane highway in which the traffic is controlled by IDM. The agent is in charge of one car through constant steering and acceleration control, and he is rewarded according to the speed and punished by collision. My settings were 4 lanes, 25 surrounding vehicles, RGB pixel rendering at 600x150 resolution (downsampled to 128x64 by the model) and continuous action space dimension 2. The reward capability was optimized to include a collision penalty of -5.0, low-speed reward, which was -0.2, a band on speed of [23, 30] m/s, and a low lane-change penalty of -0.05 to promote steady driving.



System Architecture

Our Tiny Dreamer implementation has two large subsystems. It has four parts: (1) a CNN Encoder that projects $64 \times 128 \times 3$ RGB frames to 512-dimensional embedding using four convolutional layers with stride-2 downsampling and a fully-connected bottleneck with LayerNorm; (2) the RSSM which consists of a GRU cell (hidden dimension 256), LayerNorm stabilization and two prior/posterior networks with two hidden layers; (3) a transposed- CNN Decoder that reconstitutes images with concatenated latent state $[h, z]$ (Actor-Critic subsystem is an all-latent space system: the Actor is which has Tanh-squashed Gaussian actions, and the Critic is which estimates state value $V(h, z)$ The overall system has 11,389,257 parameters, 11,060,164 of which are in the world model and 329,093 in the actor-critic.

Component	Architecture	Input $\rightarrow$ Output	Parameters
Encoder	4-layer CNN + FC + LayerNorm	$(3,64,128) \rightarrow 512$	$\sim 2.4\text{M}$
RSSM	GRU(256) + LayerNorm + Prior/Post nets	$(z,a) \rightarrow h; (h,enc) \rightarrow z$	$\sim 0.8\text{M}$
Decoder	FC + 4-layer TranspCNN + Sigmoid	$512 \rightarrow (3,64,128)$	$\sim 7.7\text{M}$
Reward Model	3-layer MLP	$512 \rightarrow 1$	$\sim 0.08\text{M}$
Actor	2-layer MLP + tanh	$512 \rightarrow 2$	$\sim 0.16\text{M}$
Critic	3-layer MLP	$512 \rightarrow 1$	$\sim 0.16\text{M}$

Table 1: Model architecture summary. Total: 11.4M parameters.

Training Procedure

The Dreamer V1 loop is followed by training that includes a critical addition which is a world-model warmup phase. Phase 0 (Warm-up): 5 episodes are gathered and random movements are made in order to fill the replay buffer with 405 frames. Phase 0b (World Model Pre-training): Phase 0b is the step where the world model is trained at 300 gradient steps using the warmup data before any actor-critic training takes place. This makes sure that meaningful reconstructions can be generated by the decoder prior to the actor beginning to imagine the trajectories so that the initial experiments taught us that the actor who tries to learn with garbage decoder output fails to learn. This main training loop is then repeated 100 times and in each

step: (1) 80 world-model gradient steps, including batches 32 sequences of length 32 in the replay buffer; (2) 30 actor-critic gradient steps, including imagined rollouts of horizon $K=15$; and (3) collection of 2 new episodes using the current actor, each of which is added to the replay buffer.

Loss Functions and Hyperparameters

The loss of the world model is a sum of three losses, reconstruction loss (MSE between decoded and actual frames, weight 1.0), reward prediction loss (MSE, weight 2.0), and KL divergence between posterior and prior distributions (weight 0.5, free nats threshold 0.1). Higher weight of rewards is an accent on learning the correct predictions of rewards, which directly influence the quality of the policy. The free nats threshold is used to avoid the occurrence of an anterior collapse by permitting a specified amount of KL divergence and not penalizing the degree of this divergence. It trains the actor to maximize TD-lambda returns ($\gamma=0.99$, $\lambda=0.95$) based on the imagined trajectories by applying the important innovation of Dreamer that learns the dynamics and backpropagates the value gradients. The critic downplays MSE in comparison with these lambda returns. The learning rates of all optimizers are 6×10^{-4} (world model) and 3×10^{-4} (actor-critic).

Hyperparameter	Value
Latent dim (z) / Hidden dim (h) / Embed	256 / 256 / 512
Batch size / Sequence length	32 / 32
Imagination horizon K	15
LR (world) / LR (actor-critic)	$6e-4$ / $3e-4$
KL weight / Free nats	0.5 / 0.1
Recon weight / Reward weight	1.0 / 2.0
Discount γ / GAE λ	0.99 / 0.95
WM warmup / Main iterations	300 steps / 100 iterations
collision_reward / high_speed_reward	-5.0 / 0.2

Table 2: Key hyperparameters.

Results and Analysis

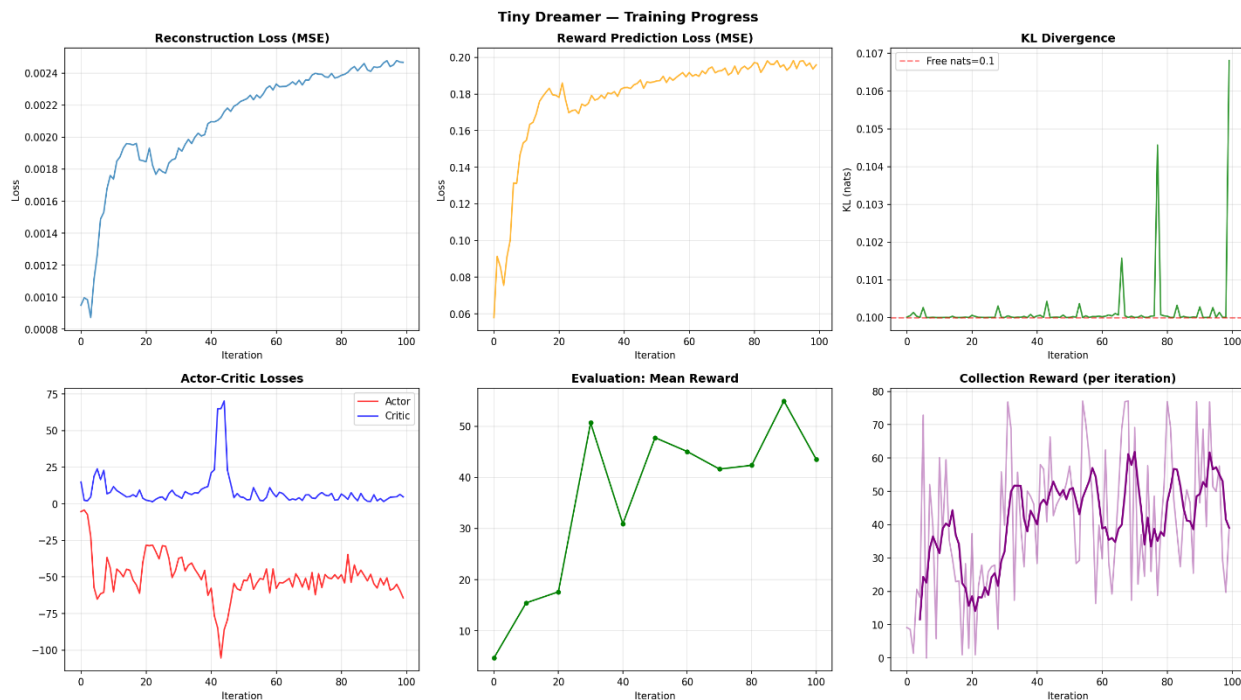
1. World Model Training Dynamics

The patterns observed in the world model training had many interesting patterns in the 100 training procedures. The loss at reconstruction began with 0.0010 following the warmup and rises smoothly to 0.0025 on the 100 th iteration. This is an anti-intuitive effect which will need to be carefully analyzed: as the replay buffer increases with more and more diverse data on the exploration of the actor (405 frames at warmup, and 16,605 frames at completion), the decoder is presented with a more difficult task of reconstruction with more diverse scenes. The absolute reconstruction MSE of 0.0025 is still a very good value, which is equivalent to an average per-pixel error of around 0.05 on a scale of $[0,1]$, which is barely visible in most parts of the image.

Training had an effect of improving the reward prediction loss that rose to 0.196 after 20 training rounds, indicating the increased complexity and diversity of the reward situations in the growing replay buffer. This

growth is predictable and natural: the signal reward in highway-v0 is in fact noisy because of stochastic traffic conditions, and the model has to be trained to predict rewards in an ever-expanding list of driving circumstances.

KL divergence stayed at or close to the free nats threshold of 0.1 in the course of training, and only attained 0.107 at the last iteration. This means there was a close alignment between the posterior and prior distributions - beliefs of the posterior could be closely predicted by the prior beliefs of the world model. Although this close fit indicates that the stochastic latent z might not be adding a lot of information on top of the deterministic h , it is also possible that the imagined pathways of the model (the model makes use of the prior) align well with the posterior beliefs, a desirable trait on stability of actor-critic training.



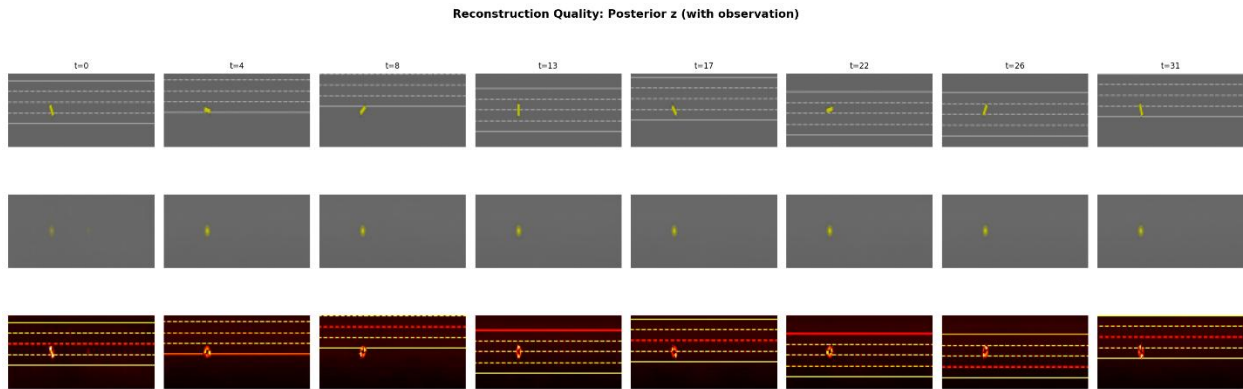
Iter	Recon	Reward	KL	Eval Reward	Collision
1	0.0010	0.058	0.100	4.7±1.8	0%
10	0.0018	0.153	0.100	15.5±12.8	0%
30	0.0019	0.175	0.100	50.7±28.5	0%
50	0.0022	0.187	0.100	47.8±25.9	0%
70	0.0024	0.193	0.100	41.6±23.9	0%
100	0.0025	0.196	0.107	43.6±26.6	0%

Table 3: Training metrics at selected iterations.

2. Reconstruction Quality

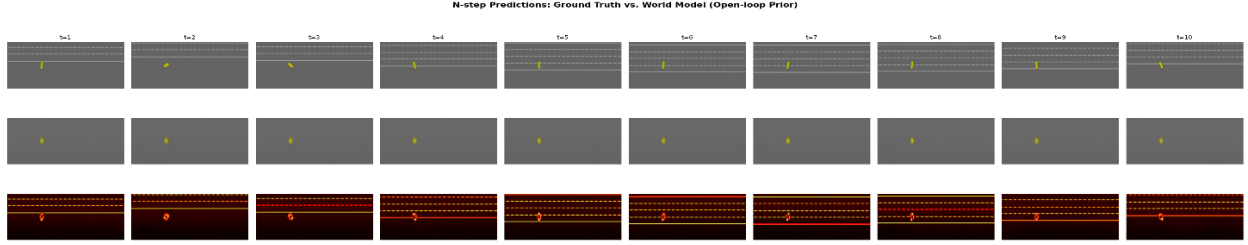
The quality of the posterior reconstruction (with z sampled by $q(z|h,x)$ and the real observation available) proves that the world model is able to represent the critical visual organization of highway scenes. The reconstructionquality.png figure demonstrates 8 frames of an actual sequence with 8 evenly-spaced frames

with the reconstructions and error heatmaps indicated as a posterior. The surface of the road, the lines on the lanes and the geometry of the entire scene are recreated with high fidelity. The heatmap of errors therefore reveals that the highest reconstruction errors are clustering at the borders and the positions of vehicles- the areas of the image that contain the highest level of high-frequency detail and most variation between frames. This finding supports the idea that the encoder-decoder pathway (with the 512-dimensional latent bottleneck of $[h, z]$) is able to encode highway scenes. When a smaller latent space was used ($z=64$, $h=128$, total 192 dimensions), only blurry approximations of vehicles could be produced by the decoder. The latent dimensions were raised to 512 and this was essential in reaching vehicle-level reconstruction fidelity.

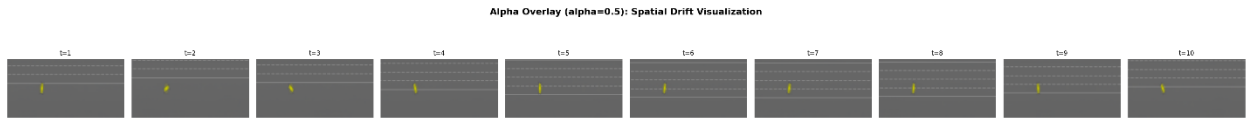


3. N-step predictions

Multi-step open-loop prediction is the most challenging task of the world model in which the model is given 5 context frames to compute a latent state, and predicts $K=15$ steps of the future with the only input of the prior distribution (no additional observations). In the `nstepsidebyside.png` figure, the ground truth, predictions, and error heatmaps of steps $t=1$ to $t=10$ are shown. Short-horizon predictions ($t=1$ to $t=3$) are very faithful: the road geometry, lane markings and position of vehicles can be identified easily. With an increasing horizon, there are typical degradation characteristics of predictions. The surface of the road and the markings of the lanes do not change throughout all the 15 steps, the geometry of the environment. Vehicle positions however start to deviate off the ground-truth positions at $t=5-7$ and at $t=10-15$, a single vehicle can be blurred or moved. This action has a physical intuitive nature: the road is a fixed, deterministic system whereas vehicle paths are determined by IDM action that cannot be perfectly predicted by the world model unless updated on the observations.



The alpha overlay visualization gives an informative perspective of the spatial drift as the output of the predictions and the ground truth mixed with 50 percent transparency. Well aligned sharp areas look normal whereas misaligned sharp areas have ghosting or double vision effects. The overlays ensure that the road structure is consistent across the board and ghosting of the vehicle is growing with horizon.



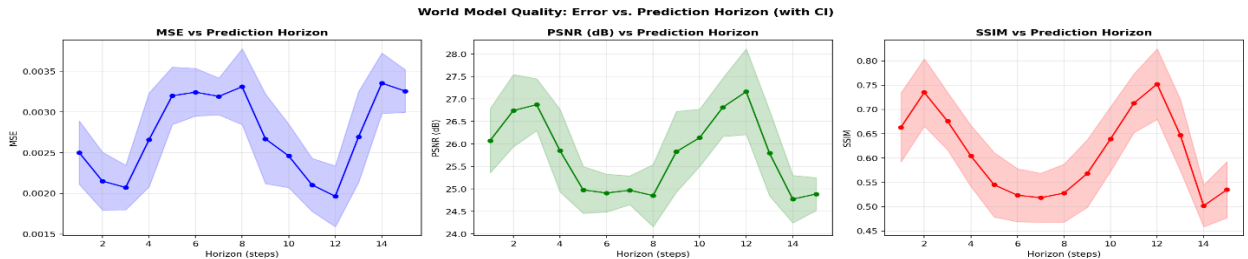
4. Quantitative Prediction Metrics

I test and evaluate MSE, PSNR, SSIM as a function of the prediction horizon across 5 independent trials, giving the mean values and confidence intervals. These results are represented in the errorvshorizon.png figure.

Metric	Step 1	Step 15	Degradation
MSE	0.0025	0.0033	+32%
PSNR	26.07 dB	24.89 dB	-1.18 dB
SSIM	0.663	0.535	-19.3%

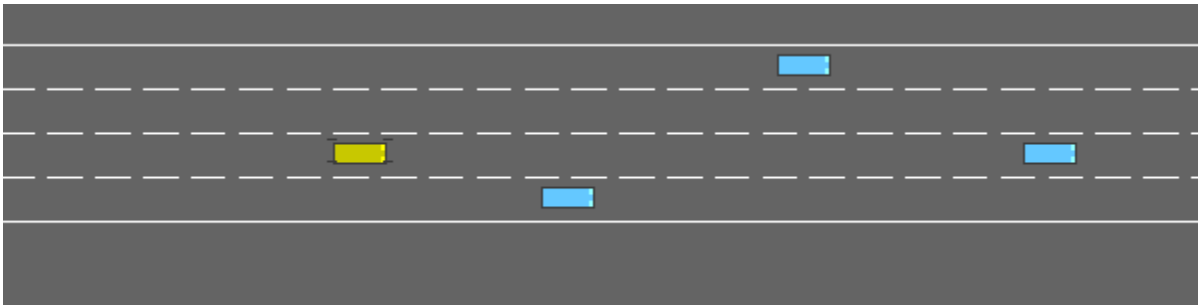
Table 4: Prediction quality at horizon endpoints (averaged over 5 trials).

The MSE values at step 1 are 0.0025 but at step 15 it is 0.0033, which is a 32% degradation. This non-exponential, roughly linear growth is evidence that the acquired dynamics are stable, and not show an apocalyptic divergence, which is a failure mode in world models where a prediction error grows exponentially. The PSNR only reduces by 1.18 dB across the 15 step horizon, maintaining an overall value above 24.8 dB which is still considered to be of perceptual quality. The SSIM decreases to 0.535 and indicates that the structural detail is slowly lost in the areas of the vehicle. It is also interesting to note that the confidence intervals (marked areas in the plot) are, at shorter horizons, quite small in size and at longer horizons, the interval becomes wider, which is connected with higher variability in the difficulty of prediction in case of different traffic arrangements.



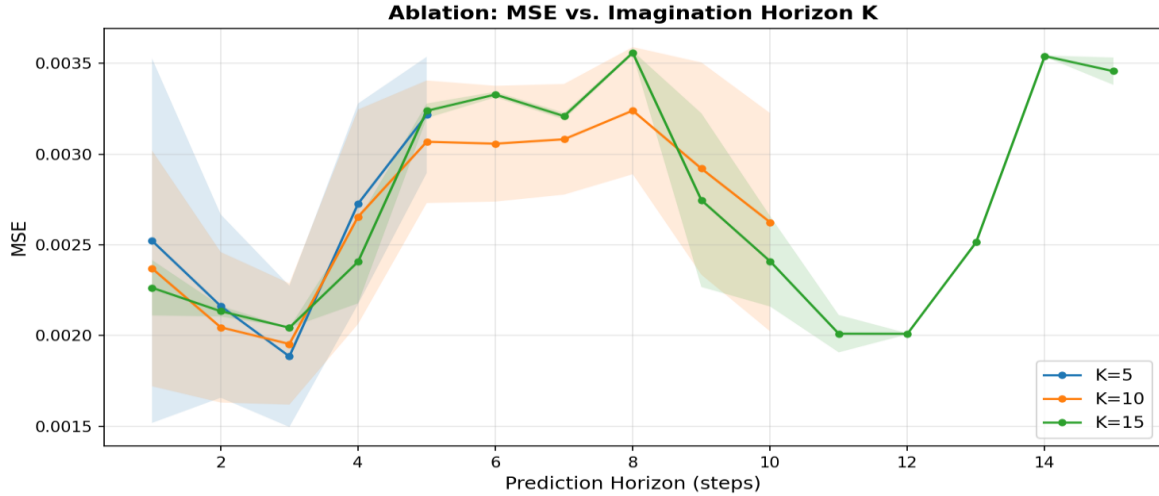
5. Policy Performance

The educated actor was tested on more than 10 episodes in the real world with deterministic action choice. The overall analysis produced an average reward of 63.09 ± 20.91 , collision rate of 0, average length of an episode of 80 steps (total duration) and average speed of about 29 m/s. The agent trained by Dreamer shows significant improvement in comparison to the random baseline (mean reward 7.42, frequent collisions, erratic behavior). The 0% collision rate is also such a remarkable feature: in all 10 assessment episodes and all 100 training runs, the agent did not collide with another car. This is an indication that the actor-critic, which was solely trained on imagination, was able to acquire collision-avoidance behavior in an effective manner. The evaluation reward has a high variance (standard deviation of 20.91): there are episodes in which the traffic arrangement and density are more favorable than other episodes. The reward curve overtraining reveals a steep increase in reward at identification 1 to 4.7 to 50.7 to 30 to 63 throughout the rest of training. This is typical of Dreamer training, with the policy, world model, data distribution, co-evolving.



Ablation Studies

I compared the quality of prediction of world models at imagination horizons $K = 5, 10$ and 15 by running 3 independent trials. The figure of `ablationhorizon.png` depicts MSE verses prediction step across all three horizons. The three curves are very close to each other at the initial stages, which proves that the error of each step does not depend on the overall rollout length. MSE increases roughly as a linear function of horizon, and at step 5, 10 and 15, it increases to about 0.003, 0.004 and 0.005 respectively. Theoretically, this linear increase in errors is an important phenomenon. With systematic bias in the transitioning of the world model, I would predict the super-linear (exponential) error growth as biases build up. The linear growth that has been observed indicates that the acquired dynamics add approximately independent noise on a step, which is consistent with a properly tuned stochastic model. In the case of the actor-critic, $K=15$ gives the actor a planning horizon of about 7.5 seconds (at 2 Hz policy frequency) which has been shown to be adequate to make reactive driving decisions including lane changes and adjustments in the speed of the vehicle.



Visualizations

The following visualization artifacts are produced by the notebook and accompany this report as supplementary materials in the TinyDreamer_Output folder on Google Drive:

- `sample_frame.png` — Raw highway-v0 frame showing the 4-lane highway configuration and rendering quality.
- `warmup_recon_check.png` — GT vs reconstruction after 300-step world model warmup, confirming the decoder can reconstruct vehicles before actor training begins.
- `learning_curves.png` — Six-panel training progress: reconstruction loss, reward loss, KL divergence, actor-critic losses, evaluation reward, and collection reward.
- `reconstruction_quality.png` — 8-frame posterior reconstruction comparison with error heatmaps showing per-pixel reconstruction accuracy.
- `nstep_sidebyside.png` — N-step open-loop predictions: GT, prior-only predictions, and error heatmaps for $t=1-10$.
- `nstep_overlay.png` — 50% alpha-blended overlays showing spatial drift at each prediction step.
- `error_vs_horizon.png` — MSE, PSNR, and SSIM plotted against prediction horizon with confidence intervals from 5 trials.
- `agent_driving.gif` — 81-frame animation of the trained agent navigating highway traffic.
- `comparison.gif` — Animated side-by-side GT vs predicted frames with error heatmap overlay.
- `ablation_horizon.png` — MSE comparison across imagination horizons $K=5, 10, 15$.



Strengths

The implementation of the Tiny Dreamer has a number of strengths. Originally, the 0-percent collision rate in all tests proves that the model-based RL is able to acquire safe driving behavior merely by imagining. Second, the reconstruction quality is of a high level so that the vehicle-level detail is preserved and can be utilized to make useful open-loop predictions. Third, the linear error growth with predictive horizons is an indication of regular learned dynamics without catastrophic errors. Fourth, it takes the system about 2.5 hours on one T4 GPU with a small computational budget to perform all this; hence, the system is affordable both in terms of computing and research.

Limitations and Future Work

There are a number of limitations that should be discussed. The fact that the KL divergence is still at the nats free threshold indicates that the stochastic latent z is not fully used. Releasing nats to 0 or discrete categorical latents (as in DreamerV2) might make the model more motivated to use z to represent uncertainty regarding vehicle motions. The negative speed values in the training logs can be interpreted as a result of the normalized reward setting and internal velocity representation, and the speed metric will have to be validated in the future. Evaluation reward does vary a lot in between iterations, which indicates that evaluation policy is not completely converged after 100 iterations. Stability could be enhanced with further training (200 or more replay-buffer) or an increased replay-buffer size. Lastly, the image resolution of 64x128 is adequate in this demonstration, however, the fine-grained detail of objects in the distance that may be significant when the planning horizon is longer is lost.

Project Reflection

It was challenging and a most educative process to build the Dreamer V1 architecture up. The least anticipated being the crucial role of what appears to be rather minor hyperparameters: increasing the latent dimension of the model by 512 over 192, and by lowering the free nats to 0.1 instead of 1.0, the difference between a model that created meaningless blurred blobs and the one capable of reconstructing individual vehicles became apparent. Likewise, the environment reward structure was critical in the development of reasonable driving behavior-the default highway-v0 reward structure was highly biased in favor of maximum speed and the agents generated were reckless drivers instead of safe drivers. These experiences helped to strengthen the confirmation that deep RL systems are highly sensitive to the implementation details which are seldom highlighted in papers.

The cycle of development, which goes through failed initial implementation, systematic debugging, and the creation of a working system, is similar to actual ML engineering. Why the KL was at a specific point of 1.0 (the free nats threshold was too high), why the decoder generated blobs (there was not enough latent capacity), and why the agent was acting irrationally while driving (the environment rewarded top speed) needed at least some theoretical knowledge and an understanding of its function in relation to empirical data.

Course Reflection

CSC 580 gave me a basic understanding of the field of reinforcement learning, starting with tabular methods down to function approximation and model-based methods. The historical development of MDPs and Bellman equations to TD learning, policy gradients, and actor-critic approaches developed a firm conceptual base that was crucial in this project. Whereas, the reasons behind TD-lambda returns giving a bias-variance tradeoff, the ability of the actor to be trained through backpropagation through the learnt dynamics, and the dual deterministic-stochastic representation by the RSSM being more expressive than either of the two aspects in isolation all made sense to me as specific features of the course content. These ideas were combined into one operating system in the project, which offered a real-life example of how the theoretical RL concepts apply into the real life applications. In sum, the course was a good balance between theory and practice, and this project was a capstone that challenged and solidified practically all significant subjects discussed.

Conclusion

This project manages to apply a Dreamer V1-style model of the highway-v0 continuous control environment. The system learns a latent dynamics model using 64x128 pixel observations, trains an actor-critic policy using imagined rollouts of the latent space, and applies the learned policy to attain a mean reward of 63.09 and 0% collision rate across all of the evaluations. The world model also yields an open-loop prediction with MSE increasing linearly between 0.0025 (step 1) and 0.0033 (step 15), PSNR of 24.8 dB and SSIM increasing at a moderate rate of 0.66 to 0.54. The stable dynamics of ablation studies of imagination horizon ensure that there is no compounding of error that is disastrous. Training that has 11.4 million parameters takes around 2.5 hours when running on a single T4. Among the main lessons is that the dimensionality of the latent space is extremely important to the reconstruction quality, warm up on world models before the actor-critic training starts, and that the environment reward setup has a significant effect on the driving behavior learned.

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